

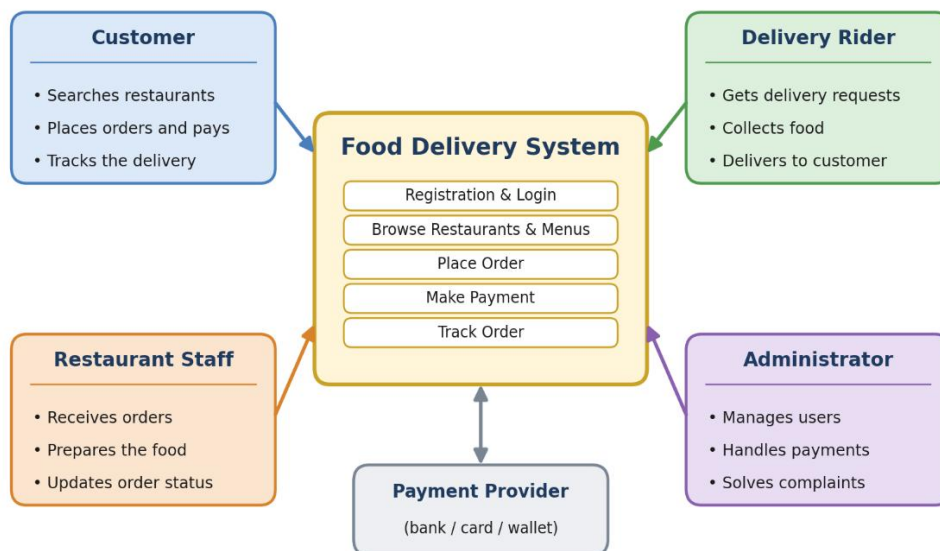
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BS APPLIED COMPUTING (B)

SOFTWARE ENGINEERING

Github link : <https://github.com/ahsankhen52-jpg/Software-Engineering>

Topic: Food Delivery System

Figure 1: System Overview and Main Users



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Purpose

A food delivery system is made so that people can order food online without going to the restaurant or calling them. The customer opens the app, looks at the restaurants and their menus, picks the food, places the order and gets it at home. The restaurant receives the order on its screen and starts preparing it, and then a delivery rider picks it up and brings it to the customer. Because all of this happens on one platform, orders are completed faster and there are fewer mistakes than when orders are taken on the phone. This also saves the people from ordering by standing in the lines.

It is also useful for restaurants because they can reach more people, and for riders because they get regular delivery work.

Users

The main users of a Food Delivery System are:

1. **Customers:** They search for restaurants, choose food, place orders, make payments and track their orders. After delivery they can also give a rating.
2. **Restaurant Staff:** They see the incoming orders, accept them, prepare the food and change the order status. They can also update the menu and prices.
3. **Delivery Riders:** They get delivery requests, collect the food from the restaurant and take it to the customer's home.
4. **Administrators:** They look after the whole system, like user accounts, restaurants, orders, payments and complaints.

Stakeholders

Stakeholders are everyone who is affected by the system or who cares about how it works, even if they do not use it themselves. The main stakeholders are:

1. **Customers:** They want to order food easily and get it on time without any problem.
2. **Restaurant Owners:** They want more orders and more customers so that their business can grow.
3. **Delivery Riders:** They depend on the system to get delivery jobs and earn money.
4. **Food Delivery Company:** It runs the platform. It wants the service to work well, the customers to stay happy and the business to make a profit.
5. **Payment Service Providers:** Banks and mobile wallet companies process the payments. They need the payments to be safe and quick.
6. **Government and Regulators:** They want the company to follow the rules about food safety, taxes and keeping people's data private.
7. **Developers and System Administrators:** They build the system, keep it running and fix the problems when they come.

Functional Requirements

1. **User Registration and Login:** Customers, restaurant staff and riders should be able to make an account and log in safely. A phone number or email can be used for verification, and there should be a way to reset a forgotten password. Each type of user should only see the pages meant for them.
2. **Browse Restaurants and Food:** Customers should be able to search for restaurants and see their menus, food items, prices, ratings and whether an item is available. Filters like food , price and rating make the search easier.

- 3. Place an Order:** The customer selects items, adds them to the cart, can change the cart, writes the delivery address and places the order. The restaurant should get the order as soon as it is confirmed.
- 4. Make Payment:** Customers can pay by cash on delivery, card or mobile wallet. The system should show the total bill clearly and give a receipt. If an order is cancelled or a payment fails, the money should be returned.
- 5. Track Order:** Customers should be able to see the status of their order, like Order Placed, Preparing, On the Way and Delivered. They should also get a notification when the status changes.

Non-Functional Requirements

- 1. Performance:** The system should be fast. Searching and opening a menu should not take more than a few seconds, and placing an order should also be quick, even when many users are online.
- 2. Security:** Passwords, personal information and payment details must be protected from people who should not see them. This can be done with secure connections, encrypted passwords and a trusted payment gateway.
- 3. Availability:** The system should be working almost all the time, especially at lunch and dinner time when most people order. Backups should be kept so the data can be recovered if something breaks.
- 4. Scalability:** When the number of customers, restaurants and orders grows, the system should still work properly. It should be able to handle busy times like weekends and festivals.
- 5. Usability:** The app should be simple. A new user should be able to place an order in a few steps without anyone teaching them. It should also work well on a mobile phone.

Risks

- 1. Payment Failure:** An online payment can fail because of a weak internet connection, a bank problem or wrong card details. Sometimes the money is cut but the order is not placed. This makes customers angry and creates complaints. To handle it, the system should use a reliable payment gateway, show clear error messages and refund the money quickly.
- 2. Delivery Delays:** Delivery can be late because of traffic, bad weather, a wrong address or too few riders. Late delivery means cold food, and customers may stop using the app. The system can help by giving a realistic delivery time, checking the address and keeping enough riders available at busy times.
- 3. Data Security Problems:** If the system is not secured properly, hackers can steal the personal and payment information of users. This can cause financial loss, legal trouble and a bad name for the company. Regular security testing, encryption and limited access to data can reduce this risk.

Quality Attributes

1. **Correctness:** The software should give the right result. The prices, taxes, delivery charges and final bill must be calculated correctly, and the restaurant should get exactly what the customer ordered.
2. **Reliability:** The system should work well every time and should not fail randomly. Orders should not get lost or changed, even when a lot of people are ordering together.
3. **Maintainability:** It should be easy for developers to fix bugs or add new things, like a new payment method, without breaking the rest of the system. Clean code, testing and documentation help with this.
4. **Platform Rating:** The platform that are used by customers for food delivery must be having 5 star rating.

Where AI Could Be Used

1. **Food Recommendations:** AI can look at what a customer ordered or searched before and suggest restaurants and food they may like. This saves the customer's time and can increase sales for the restaurants.
2. **Delivery Route Planning:** AI can check traffic and road conditions and show the rider a faster route. It can also give a better estimate of the delivery time.
3. **Customer Support:** An AI chatbot can answer common questions about orders, payments, delivery time and cancellations at any time of the day. Support staff then only need to handle the difficult cases.
4. **Predicting Food Demand:** AI can guess when there will be more orders, for example on a Friday night. Restaurants can then prepare enough food and the company can arrange enough riders.
5. **Fraud Detection:** AI can notice strange activity such as fake orders or someone using a stolen card, and warn the company early.

What Happened if AI Is Used Incorrectly

1. **Wrong Recommendations:** If the AI does not understand what a customer likes, it will suggest food they do not want. It can be more serious if a customer has an allergy and the AI suggests something that is not suitable for them.
2. **Incorrect Customer Support:** The chatbot may give a wrong answer about an order, refund or cancellation. The customer may believe it and face a loss, and then it is not clear who is responsible.
3. **Privacy Issues:** The AI uses customer data like names, addresses and phone numbers. If this data is not handled carefully, it can leak or be misused.

4. **Wrong Predictions:** If the demand prediction is wrong, restaurants can waste food or there may not be enough riders. A mistake in fraud detection can also block a genuine customer.

These problems can be reduced by testing the AI properly before using it, checking its results regularly and keeping customer data safe. For important matters like money and refunds, a human staff member should be able to step in.

SUMMARY :

A Food Delivery System makes ordering food easier by connecting customers, restaurants and delivery riders on one platform. Customers can search for food, place orders, pay and track their delivery, restaurants can manage their orders, and riders can handle the deliveries. The system needs clear requirements and good quality, and it has to deal with risks like payment failure, late delivery and data theft.

AI can make the system more useful through food recommendations, route planning, customer support, demand prediction and fraud detection. But it has to be used carefully, because wrong information or careless use of customer data can create big problems. So AI does not replace software engineering, it actually makes good engineering more necessary. Because AI generates content and software engineers ensure the quality and reliability of the whole system.